Lesson 1: Can This Data Be Trusted?

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students learn how scientists judge whether data can be trusted. They are introduced to uncertainty, precision, accuracy, resolution of measurement, random error, and systematic error before they begin planning and presenting investigations in the rest of the unit.

Why this lesson comes first

This lesson prepares students for the practical and data-handling skills used throughout Lessons 2-8. Before students choose graph types, calculate means, draw best-fit lines, or evaluate anomalous results, they need to understand why measurements are never perfect and why repeated readings help us judge whether results are repeatable.

Learning objectives

By the end of the lesson, students should be able to:

1. Explain why measurements always have some uncertainty.
2. Calculate uncertainty from repeated readings.
3. Distinguish between precision and accuracy.
4. Explain what is meant by the resolution of a measuring instrument.
5. Describe random errors and systematic errors.
6. Suggest ways to make data more repeatable.

Key vocabulary

Keyword	Student-friendly definition
Repeatable	The same person, method and equipment give similar results again
Reproducible	Another person, or a different method or equipment, gives similar results
Uncertainty	The interval within which the true value can be expected to lie
Range	The maximum and minimum values of the independent or dependent variables
Precision	How close repeated measurements are to each other
Accuracy	How close a measurement is to the true value
Resolution	The smallest change a measuring instrument can detect
Random error	Unpredictable variation between repeated measurements
Systematic error	An error that affects all results in the same way
Mean	The average value

AQA-style key ideas

Idea	What students need to understand
Uncertainty	Whenever a measurement is made, there is always some uncertainty about the result obtained
Estimating uncertainty from repeats	Use the range of repeated measurements about the mean as a measure of uncertainty
Range	The maximum and minimum values of the independent or dependent variables, for example "from 10 cm to 50 cm"; it is not a single number
Simple uncertainty calculation	Uncertainty = (maximum value - minimum value) / 2
Precision	Precise measurements are close together
Accuracy	Accurate measurements are close to the true value
Random error	Random errors cause readings to vary unpredictably
Systematic error	Systematic errors shift readings in the same direction each time
Resolution	A measuring instrument with smaller divisions has better resolution

Equipment and resources

- Mini whiteboards or exercise books
- Projected examples of repeated readings
- Meter rulers
- Stopwatches
- A toy car, trolley, or ball
- Optional: different measuring instruments, for example a ruler marked in cm and a ruler marked in mm
- Optional: printed worksheet with data sets

Starter: Which data set would you trust most?

Show students three sets of repeated measurements for the same experiment.

Example question:

> *A student measures the time taken for a toy car to travel down a ramp.*

Data set	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
A	1.20	1.23	1.19	1.21
B	1.05	1.47	1.12	1.21
C	0.90	0.93	0.89	0.91

Ask students:

1. Which set of readings is most precise?
2. Which set has the biggest spread?
3. Which set might be inaccurate even though it is precise?
4. Why is it useful to repeat measurements?

Expected discussion:

- Data set A is precise because the readings are close together.
- Data set B is less precise because the readings are spread out.
- Data set C is precise, but could be inaccurate if the true fall time is closer to 1.21 s.
- Repeats help us spot variation and calculate a mean that is closer to the true value.

Main teaching: Precision and accuracy

Introduce the difference between precision and accuracy.

Situation	Meaning
Accurate and precise	Measurements are close together and close to the true value
Precise but not accurate	Measurements are close together but far from the true value
Accurate on average but not precise	Measurements are spread out, but the mean is close to the true value
Neither accurate nor precise	Measurements are spread out and far from the true value

Useful wording:

> *Precision is about how close repeated measurements are to each other.*
> *Accuracy is about how close a measurement is to the true value.*

Main teaching: Random and systematic errors

Introduce the two main types of error.

Error type	Meaning	Example
Random error	Unpredictable variation between repeats	Different reaction times when starting or stopping a stopwatch
Systematic error	A repeated error that shifts all readings in the same direction	A ruler starts from the wrong position or a balance is not set to zero

Students should understand:

- Repeating measurements and calculating a mean can reduce the effect of random errors.
- Repeating measurements does not remove systematic errors.
- Systematic errors are improved by checking equipment and method.

Main teaching: Resolution of measurement

Resolution means the smallest change a measuring instrument can detect.

Examples:

Instrument	Resolution
Ruler marked every 1 cm	1 cm
Ruler marked every 1 mm	1 mm
Stopwatch reading to 0.01 s	0.01 s
Balance reading to 0.1 g	0.1 g

Key point:

> *A smaller resolution allows more precise measurements, but it does not automatically make the results accurate.*

Activity 1: Calculate uncertainty from repeated readings

Teach students to estimate uncertainty using:

> *Uncertainty = (maximum value - minimum value) / 2*

Example:

Trial	Time / s
1	1.20
2	1.25
3	1.21

Step-by-step:

1. Maximum value = 1.25 s
2. Minimum value = 1.20 s
3. Range: from 1.20 s to 1.25 s
4. Uncertainty = (1.25 - 1.20) / 2 = 0.025 s

The result can be written as:

> *Mean time = 1.22 s +/- 0.025 s*

For Year 9, students can also round this sensibly:

> *Mean time = 1.22 s +/- 0.03 s*

Activity 2: Practice questions

Students calculate the mean, state the range (the maximum and minimum values) and calculate the uncertainty for each data set.

Data set	Trial 1	Trial 2	Trial 3	Mean	Range	Uncertainty
A	12.0	12.4	12.1			
B	8.4	9.1	8.6			
C	24.9	25.1	25.1			
D	3.20	3.65	3.25			

Challenge questions:

1. Which data set is most precise?
2. Which data set has the largest uncertainty?
3. Which data set might need to be repeated?
4. What could cause the readings to be spread out?

Activity 3: Mini practical

Students complete a short repeated-measurement task.

Possible options:

1. Measure reaction time using a dropped ruler.
2. Measure the distance travelled by a toy car after rolling down a small ramp.
3. Measure the bounce height of a ball.
4. Measure the length of the same object using different rulers.

Students collect at least 3 repeated readings.

Measurement	Trial 1	Trial 2	Trial 3	Mean	Range	Uncertainty

Students then answer:

1. Were your results precise?
2. What was your uncertainty?
3. What random errors may have affected your results?
4. What systematic errors may have affected your results?
5. How could you make your data more repeatable?

Check for understanding

Ask students:

1. What does precision mean?
2. What does accuracy mean?
3. What is the resolution of a ruler marked every 1 mm?
4. Why can a result be precise but not accurate?
5. What is a random error?
6. What is a systematic error?
7. Why does taking repeats help with random error?
8. Why does taking repeats not fix systematic error?

Plenary: Exit ticket

Students complete:

1. A precise set of results is _____.
2. An accurate result is _____.
3. The uncertainty is calculated using _____.
4. One possible random error was _____.
5. One possible systematic error was _____.
6. To make my data more repeatable, I could _____.

Possible prep

Students are given three small data sets and must:

1. Calculate the mean.

2. State the range (the maximum and minimum values).
3. Calculate the uncertainty using (largest - smallest) / 2.
4. Identify which data set is most precise.
5. Suggest one possible random error and one possible systematic error.

Teacher notes

This lesson should stay practical and concrete. The aim is not to teach advanced uncertainty analysis, but to give students the language and habits they will need for the rest of the unit.

The most important links to later lessons are:

1. Repeats help students judge whether results are repeatable.
2. Means are useful because repeated measurements vary.
3. Unusual results may be anomalous results, but should be checked before being ignored.
4. Graphs and conclusions are only as good as the quality of the data behind them.
5. Evaluations should mention specific random errors, systematic errors, and realistic improvements.

Common misconception:

Students often think precise means correct. Emphasize that precise measurements are close together, but they may still be inaccurate if there is a systematic error.